HW#3: required reading S. Blundell, Ch.5

Order and magnetic structures

5

In the previous chapter the different types of magnetic interaction which operate between magnetic moments in a solid have been presented. In this chapter we will consider the different types of magnetic ground state which can be produced by these interactions. Some of these ground states are illustrated in Fig. 5.1. The different ground states include ferromagnets in which all the magnetic moments are in parallel alignment, antiferromagnets in which adjacent magnetic moments lie in antiparallel alignment, spiral and helical structures in which the direction of the magnetic moment precesses around a cone or a circle as one moves from one site to the next, and spin glasses in which the magnetic moments lie in frozen random arrangements. This chapter will be concerned with showing how, in broad terms, the interactions discussed in the previous chapter lead to these differing ground states. In the following chapter the phenomenon of order will be examined in a more general context and it will be seen that order is a consequence of broken symmetry.

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Notation reminder: In this book J refers to the exchange constant, J is total angular momentum.

5.1 Ferromagnetism

A ferromagnet has a spontaneous magnetization even in the absence of an applied field. All the magnetic moments lie along a single unique direction.[1] This effect is generally due to exchange interactions which were described in the previous chapter. For a ferromagnet in an applied magnetic field $\mathbf{B}$, the appropriate Hamiltonian to solve is

$$\hat{\mathcal{H}} = -\sum_{ij} \mathsf{J}_{ij}\mathbf{S}_i \cdot \mathbf{S}_j + g\mu_{\mathrm{B}} \sum_j \mathbf{S}_j \cdot \mathbf{B}, \tag{5.1}$$

and the exchange constants for nearest neighbours will be positive in this case, to ensure ferromagnetic alignment. The first term on the right is the Heisenberg exchange energy (see eqn 4.7). The second term on the right is the Zeeman energy (see eqn 1.35). To keep things simple to begin with, let us assume[2] that we are dealing with a system in which there is no orbital angular momentum, so that $L = 0$ and $J = S$.

[1] In fact in many ferromagnetic samples this is not true throughout the sample because of domains. In each domain there is a uniform magnetization, but the magnetization of each domain points in a different direction from its neighbours. See Section 6.7 for more on magnetic domains.

[2] We will relax this assumption later in Section 5.1.4.

5.1.1 The Weiss model of a ferromagnet

To make progress with solving eqn 5.1 it is necessary to make an approximation. We define an effective molecular field at the ith site by

$$\mathbf{B}_{\mathrm{mf}} = -\frac{2}{g\mu_{\mathrm{B}}} \sum_j \mathsf{J}_{ij}\mathbf{S}_j. \tag{5.2}$$

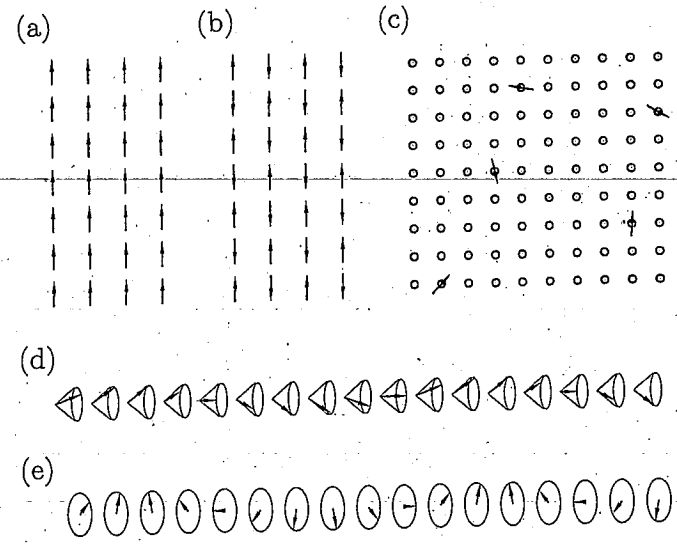

Fig. 5.1 Various spin arrangements in ordered systems: (a) ferromagnets, (b) antiferromagnets, (c) spin glasses and (d) spiral and (e) helical structures.

Now focus on the ith spin. Its energy is due to a Zeeman part $g\mu_{\text{B}}\mathbf{S}_i \cdot \mathbf{B}$ and an exchange part. The total exchange interaction between the ith spin and its neighbours is $-2\sum_j \mathrm{J}_{ij}\mathbf{S}_i \cdot \mathbf{S}_j$, where the factor of 2 is because of the double counting.[3] This term can be written as

$$-2\mathbf{S}_i \cdot \sum_j \mathrm{J}_{ij}\mathbf{S}_j = -g\mu_{\text{B}}\mathbf{S}_i \cdot \mathbf{B}_{\text{mf}}. \tag{5.3}$$

[3] See Section 4.2.1.

Hence the exchange interaction is replaced by an effective molecular field $\mathbf{B}_{\text{mf}}$ produced by the neighbouring spins. The effective Hamiltonian can now be written as

$$\hat{\mathcal{H}} = g\mu_{\text{B}} \sum_i \mathbf{S}_i \cdot (\mathbf{B} + \mathbf{B}_{\text{mf}}) \tag{5.4}$$

which now looks like the Hamiltonian for a paramagnet in a magnetic field $\mathbf{B} + \mathbf{B}_{\text{mf}}$. The assumption underpinning this approach is that all magnetic ions experience the same molecular field. This may be rather questionable, particularly at temperatures close to a magnetic phase transition, as will be discussed in the following chapter. For a ferromagnet the molecular field will act so as to align neighbouring magnetic moments. This is because the dominant exchange interactions are positive. (For an antiferromagnet, they will be negative.)

Since the molecular field measures the effect of the ordering of the system, one can assume that

$$\mathbf{B}_{\text{mf}} = \lambda \mathbf{M} \tag{5.5}$$

where λ is a constant which parametrizes the strength of the molecular field as a function of the magnetization. For a ferromagnet, $\lambda > 0$. Because of the large Coulomb energy involved in the exchange interaction, the molecular field is often found to be extremely large in ferromagnets.

We are now able to treat this problem as if the system were a simple paramagnet placed in a magnetic field $\mathbf{B} + \mathbf{B}_{\text{mf}}$. At low temperature, the moments can be aligned by the internal molecular field, even without any applied field

being present. Notice that the alignment of these magnetic moments gives rise to the internal molecular field that causes the alignment in the first place, so that this is something of a 'chicken-and-egg' scenario. At low temperature the magnetic order is self-sustaining. As the temperature is raised, thermal fluctuations begin to progressively destroy the magnetization and at a critical temperature the order will be destroyed. This model is known as the Weiss model of ferromagnetism.

Pierre Weiss (1865–1946)

To find solutions to this model, it is necessary to solve simultaneously the equations

$$\frac{M}{M_s} = \mathrm{B}_J(y) \tag{5.6}$$

(see eqn 2.38) and

Reminder: we are assuming $J = S$ and $L = 0$ at this stage.

$$y = \frac{g_J \mu_B J(B + \lambda M)}{k_B T} \tag{5.7}$$

(see eqn 2.37). Without the λM term due to the molecular field, this would be identical to our treatment of a paramagnet in Section 2.4.3.

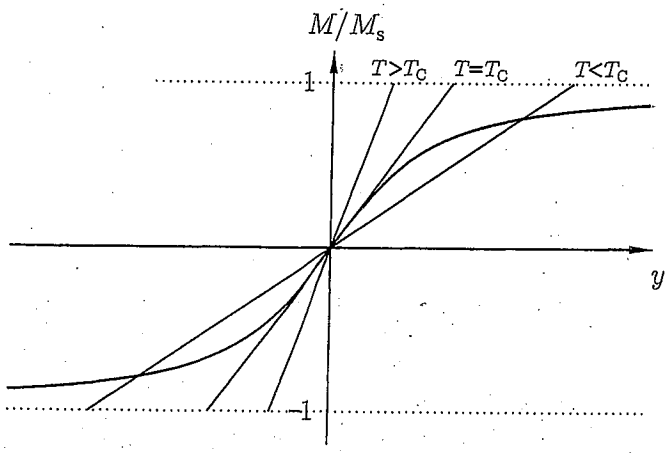

Fig. 5.2 The graphical solution of eqns 5.6 and 5.7 for $B = 0$.

These equations can be solved graphically. First, we restrict our attention to the case of $B = 0$, so that $M = k_B T y / g_J \mu_B J \lambda$. Hence the straight line produced by plotting M against y has a gradient which is proportional to temperature T as illustrated in Fig. 5.2. For high temperature, there is no simultaneous solution of eqns 5.6 and 5.7 except at the origin where $y = 0$ and $M_s = 0$. This situation changes when the gradient of the line is less than that of the Brillouin function at the origin. At low temperatures there are then three solutions, one at $M_s = 0$ and another two for M_s at $\pm$ some non-zero value. It turns out that when the curve is less steep than the Brillouin function at the origin, the non-zero solutions are stable and the zero-solution is unstable. (If the system has $M_s = 0$ for $T < T_C$, any fluctuation, no matter how small, will cause the system to turn into either one of the two stable states.) Thus below a certain temperature, non-zero magnetization occurs and this grows as the material is cooled. The substance thus becomes magnetized, even in the absence of an external field. This spontaneous magnetization is the characteristic of ferromagnetism.

The temperature at which the transition occurs can be obtained by finding when the gradients of the line $M = k_B T y / g_J \mu_B J \lambda M_s$ and the curve $M = M_s \mathrm{B}_J(y)$ are equal at the origin. For small y, $\mathrm{B}_J(y) = (J+1)y/3J + O(y^3)$.

The transition temperature, known as the Curie temperature T_C, is then defined by

$$T_C = \frac{g_J \mu_B (J+1)\lambda M_s}{3k_B} = \frac{n\lambda\mu_{eff}^2}{3k_B}. \tag{5.8}$$

The molecular field $B_{mf} = \lambda M_s$ is thus $3k_B T_C / g_J \mu_B (J+1)$ and so for a ferromagnet with $J = \frac{1}{2}$ and $T_C \sim 10^3$ K, $B_{mf} = k_B T_C / \mu_B \sim 1500$ T. This is an enormous effective magnetic field and reflects the strength of the exchange interaction.

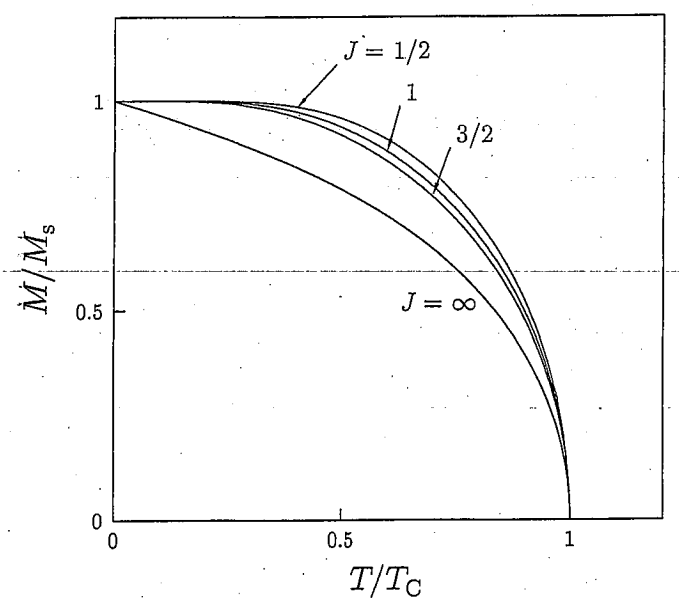

Fig. 5.3 The mean-field magnetization as a function of temperature, deduced for different values of J.

The solutions of these equations as a function of temperature are shown in Fig. 5.3 for a range of values of J. Although the form of the curves is slightly different in each case, some general features persist. The magnetization is zero for temperatures $T \geq T_C$ and is non-zero for $T < T_C$. The magnetization is continuous at $T = T_C$, but its gradient is not. This classifies the phase transition between the non-magnetic and ferromagnetic phases in this molecular field model as a second-order phase transition. The order of a phase transition is the order of the lowest differential of the free energy which shows a discontinuity at the transition. A first-order phase transition would have a discontinuous jump in the first derivative of the free energy, i.e. in quantities like the volume, entropy or the magnetization. The jump in the entropy gives a latent heat. A second-order phase transition has a discontinuity in the second derivative of the free energy, i.e. in quantities like the compressibility or the heat capacity. In the present case the discontinuity is in the gradient of the magnetization, i.e. in the second derivative of the free energy, so the transition is second order. Phase transitions and critical exponents will be considered in more detail in Section 6.4. The properties of some common ferromagnets are listed in Table 5.1.

Table 5.1 Properties of some common ferromagnets.

Material	T_C (K)	magnetic moment (μ_B /formula unit)
Fe	1043	2.22
Co	1394	1.715
Ni	631	0.605
Gd	289	7.5
MnSb	587	3.5
EuO	70	6.9
EuS	16.5	6.9

5.1.2 Magnetic susceptibility

Applying a small B field at $T \geq T_C$ will lead to a small magnetization, so that the $y \ll 1$ approximation for the Brillouin function can be used. Thus

$$\frac{M}{M_s} \approx \frac{g_J \mu_B (J+1)}{3k_B} \left(\frac{B + \lambda M}{T} \right) \tag{5.9}$$

so that

$$\frac{M}{M_s} \approx \frac{T_C}{\lambda M_s} \left(\frac{B + \lambda M}{T} \right). \tag{5.10}$$

This can be rearranged to give

$$\frac{M}{M_s} \left(1 - \frac{T_C}{T} \right) \approx \frac{T_C B}{\lambda M_s} \tag{5.11}$$

so that

$$\chi = \lim_{B \to 0} \frac{\mu_0 M}{B} \propto \frac{1}{T - T_C} \tag{5.12}$$

which is known as the Curie Weiss law.

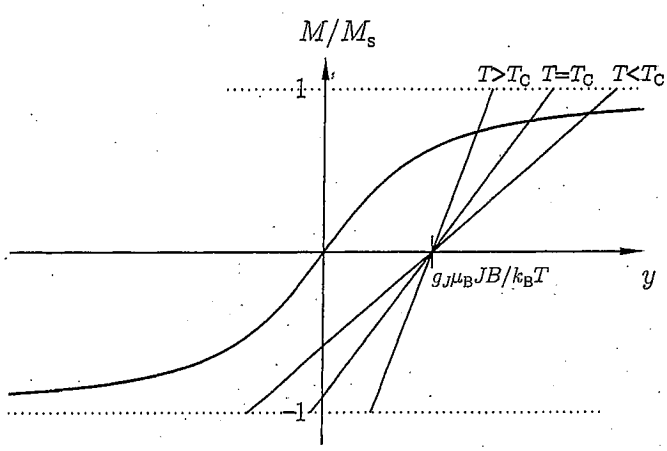

Fig. 5.4 The graphical solution of eqns 5.6 and 5.7 for $B \neq 0$.

5.1.3 The effect of a magnetic field

The effect of adding a magnetic field is to shift to the right the straight line in the graphical solution of the equations (see Fig. 5.4). This results in a solution with $M \neq 0$ for all temperatures and so the phase transition is removed. For ferromagnets in a non-zero magnetic field there is always an energetic advantage to have a non-zero magnetization with the moments lining up along the magnetic field. This removal of the phase transition can be seen in Fig. 5.5 which shows graphical solutions to eqns 5.6 and 5.7 for a range of magnetic fields. In this model it is not necessary to consider the effect of applying a magnetic field in different directions. Whichever direction the magnetic field is applied, the magnetization will rotate round to follow it. The model contains no special direction associated with the ferromagnet itself. In a real ferromagnet this is not the case, and the effect of magnetic anisotropy associated with the material will need to be considered (see the following chapter).

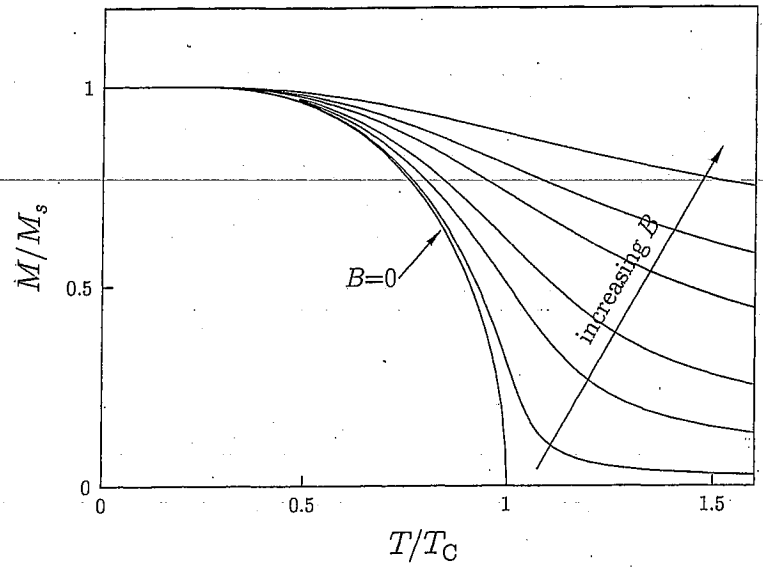

Fig. 5.5 The mean-field magnetization as a function of temperature for $J = \frac{1}{2}$, calculated for different values of the applied field B. The phase transition is only present when $B = 0$.

At $T = T_C$, the effect of the magnetic field is simple to work out analytically. At this temperature, the magnetization is given by $M \propto B^{1/3}$ for small magnetic fields. To prove this, it is necessary to take the next term in the Taylor expansion of $B_J(y)$. Writing $B_J(y) = (J+1)y/3J - \zeta y^3 + O(y^5)$, where ζ is a constant, we must solve simultaneously $M = M_s B_J(y)$ and

$$y = \frac{g_J \mu_B J(B + \lambda M)}{k_B T_C} = \frac{(B + \lambda M)}{\zeta M_s} \tag{5.13}$$

which yields

$$M = M_s \zeta \frac{(B + \lambda M)}{\lambda} - \zeta M_s \left(\frac{3J(B + \lambda M)}{\lambda(J+1)M_s} \right)^3 \tag{5.14}$$

and hence

$$B \propto (B + \lambda M)^3 \tag{5.15}$$

and given that $\lambda M \gg B$, the right-hand side is dominated by the M^3 term so that $M \propto B^{1/3}$.

5.1.4 Origin of the molecular field

When Weiss proposed his molecular field model in 1907 he was disappointed that the constant λ needed to be very large to agree with the large values of T_C found in nature. Considering only dipole fields, it was not possible to account for an internal field which, as discussed above, needs to be $\sim 10^3$ T to account for the Curie temperature of Fe. Thirty years later, Heisenberg showed that it was the exchange interaction, which involves large Coulomb energies, which is responsible for the large molecular field.[4]

The molecular field, parametrized by λ, can be related to the size of the exchange interaction, parametrized by J_{ij}. Assuming that the exchange interaction is effective only over the z nearest neighbours of an ion, where it takes

[4]The molecular field is a convenient fiction and one shouldn't think that magnetic fields $\sim 10^3$ T are experienced by electrons in ferromagnets. The exchange interaction is purely an electrostatic effect. When we talk about the molecular field we are pretending that the exchange interaction is actually internal magnetic field. The point is that, if it really existed and Heisenberg exchange didn't, the molecular field would have to be $\sim 10^3$ T to explain the effects we see.

HW #3

1

(5.5) Show that for $T \ll T_C$ the Weiss model of a ferromagnet leads to a magnetization M given by

$$M = M_s(1 - 2e^{-2T_C/T}). \tag{5.46}$$

Note that this is in disagreement with the experimentally determined power law behaviour for $T \ll T_C$ which follows

$$M = M_s(1 - AT^{3/2}) \tag{5.47}$$

where A is a constant. This low temperature behaviour can be explained by the effect of spin waves (see Section 6.6).

2

(5.7) Show that the Hamiltonian for the Heisenberg model

$$\hat{\mathcal{H}} = -\sum_{<ij>} J\mathbf{S}_i \cdot \mathbf{S}_j, \tag{5.51}$$

can be rewritten as

$$\hat{\mathcal{H}} = -\sum_{<ij>} J\left[S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+)\right]. \tag{5.52}$$

Hence show that for a Heisenberg ferromagnet ($J > 0$) the state $|\Phi\rangle$, which consists of spins on every site pointing up (say), is an eigenstate of the Hamiltonian and has energy E_0 given by

$$E_0 = -NS^2J. \tag{5.53}$$

Consider the Heisenberg antiferromagnet ($J < 0$) with the spins residing on two sublattices, each spin interacting only with those on the other sublattice. Show that the 'obvious' ground state, namely one with each sublattice ferromagnetically aligned but with oppositely directed sublattice magnetizations, is not an eigenstate of the Hamiltonian. This emphasizes that the Heisenberg antiferromagnet is a complex and difficult problem.